Complete ERP Model

ECC to S/4HANA — All Modules · Config Tables · Real Examples

FI	CO	MM	SD	PP	PM	QM	HR	AA	TR

How the ERP is Built · Layer by Layer · Integration · Migration Impact

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01 How an ERP is Built — The 7 Layers

Every SAP system is built bottom to top — you cannot skip a layer

Before any user can raise a purchase order or post an invoice, SAP must be **built layer by layer**. Think of it like constructing a building — the foundation must be solid before you add floors. Each layer depends on the one below it.

LAYER 7 — TRANSACTIONS	VA01 sales order · ME21N purchase order · FB60 vendor invoice · MIGO goods receipt
LAYER 6 — MASTER DATA	Material Master · Business Partner (Customer/Vendor) · Assets · Cost Centres
LAYER 5 — INTEGRATION CONFIG	CVI · Account Determination · Partner Det. · Revenue Acct Det.
LAYER 4 — MODULE CONFIG	MRP types · Payment terms · Pricing procedures · Depreciation keys
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LAYER 2 — FINANCIAL FOUNDATION	Chart of Accounts · Fiscal Year · Ledgers · Document types · Posting keys
LAYER 1 — CLIENT / SYSTEM	Client 100 · Languages · Currencies · Countries · Units of Measure

Golden Rule: Always configure bottom to top. Cannot create a Plant (Layer 3) before Company Code (Layer 2). Cannot create Material Master (Layer 6) before Plant (Layer 3). Cannot post a transaction (Layer 7) before master data (Layer 6) exists.

02 Layer 1 — Client & System Settings

The lowest building block — everything in SAP lives inside a client

A **Client** is the highest-level container in SAP. All config, master data and transactions live inside it. Most companies use: Client 100=Production, 200=Quality/Test, 300=Development.

Config Object	Transaction	Table	ECC	S/4	Example
Define Client	SCC4	T000	■	■	Client 100 — Sempra Production
Countries	SPRO	T005	■	■	US, DE, MX, CA, GB
Country Descriptions	OY01	T005T	■	■	US = United States
Languages	SPRO	T002	■	■	EN, DE, ES, FR
Currencies	OY03	TCURC	■	■	USD, EUR, MXN, CAD
Exchange Rate Types	OB07	TCURV	■	■	M=average, B=bank buying rate
Exchange Rates	OB08	TCURR	■	■	USD/EUR = 0.92 (daily updated)
Units of Measure	CUNI	T006	■	■	EA, KG, L, M3, HR, PC, SET
Factory Calendar	SCAL	TFACD	■	■	US = USA working days
Time Zones	SPRO	TTZZ	■	■	America/Los_Angeles
Number Range Objects	SNRO	NRIV	■	■	All doc/object number ranges

03 Layer 2 — Financial Foundation (FI)

Chart of Accounts, Company Code, Ledgers, ACDOCA — the accounting backbone

Chart of Accounts — The Master G/L Account List

The **Chart of Accounts (CoA)** is the master list of all G/L accounts. In S/4HANA, EVERY G/L account is automatically a cost element (CO). In ECC they were separate objects — a common source of errors.

Account Range	Category	Example Accounts	Notes
1xxxxx	Assets	100000=Cash, 113100=Bank, 140000=AR Recon, 191100=GR/IR, 310000=Fg Inventory	Balance sheet — debit normal
2xxxxx	Liabilities	210000=AP Recon, 220000=Accruals, 280000=Long-Term Debt	Balance sheet — credit normal
3xxxxx	Equity	300000=Retained Earnings, 310000=Share Capital	Balance sheet
4xxxxx	Revenue	400000=Product Revenue, 410000=Service Revenue	P&L — credit normal
5xxxxx	COGS	500000=Labour, 600000=Overhead, 893000=Cost of Goods Sold	P&L — debit normal
8xxxxx	Other Costs	800000=Admin, 810000=Selling, 850000=Depreciation	P&L

Config Object	Transaction	Table	ECC	S/4	Example / Notes
Create Chart of Accounts	OB13	T004	■	■	USCA=US Sempra, INT=International
G/L Account Master (CoA level)	FS00	SKA1	■	■	Account type, short text, P&L/BS
G/L Account Master (CoCode level)	FSS0	SKB1	■	■	Open item mgmt, sort key, field status
Account Groups (G/L)	OBD4	T077S	■	■	Assets, Liabilities, Revenue, Costs
Field Status Groups	OBC4	T004F	■	■	Which fields req/opt per account
Fiscal Year Variant	OB29	T009	■	■	K4=Jan-Dec (12 periods), V3=Apr-Mar
Assign FY to Company Code	OB37	T001	■	■	CoCode 1000 → Variant K4
Posting Periods	OB52	T010O	■	■	Period 1=Jan open, 13=special period
Document Types	OBA7	T003	■	■	SA=G/L, KR=Vendor inv, DR=Cust inv, AA=Asset
Document Number Ranges	FBN1	NRIV	■	■	Doc type SA → internal range 01
Posting Keys	OB41	TBSL	■	■	01=Cust Debit, 31=Vend Credit, 40=G/L Dr
Tolerance Groups (FI)	OBA3	T043T	■	■	Max posting difference: $5 or 0.5%
New G/L — Ledgers	FINSC_LEDGER	FINSC_LEDGER	Opt	MANDT=IFRS	0L=IFRS leading, 2L=local GAAP
Document Splitting	FAGL_SPLITTING	FAGL_SPLINFO	Opt	MANDT=Split by	Split by Profit Centre per line
Parallel Ledgers	FINSC_LEDGER	FAGL_SEGM	Opt	MANDT=IFRS+	IFRS + US GAAP simultaneously

S/4 Key Change — ACDOCA Universal Journal: In ECC, FI posted to BKPF+BSEG and CO posted to COEP — two separate tables that needed monthly reconciliation. In S/4HANA both post to ONE table: ACDOCA. Every line item carries ALL dimensions: G/L account, cost centre, profit centre, segment, material, plant, customer, vendor — simultaneously. Real-time P&L;, no batch jobs.

04 Layer 3 — Enterprise Structure

The organisational skeleton — plants, sales orgs, purchasing orgs, controlling area

Enterprise structure is the **framework everything else hangs from**. Get this wrong and every subsequent config and master data load has to be redone. Define this in week 1 of the project and lock it down.

```
CLIENT 100 ■■■ CONTROLLING AREA: 1000 (North America) → table TKA01 ■ ■■■ Links to Company Codes 1000, 2000 ■ ■■■ COMPANY CODE: 1000 (SoCal Gas) → table T001 ■ ■■■ Chart of Accounts: USCA ■ ■■■ Fiscal Year: K4 (Jan–Dec) ■ ■■■ Currency: USD ■ ■■■ PLANT: 1000 (Los Angeles) → table T001W ■ ■ ■■■ Storage Location: 0001 Main Warehouse → table T001L ■ ■ ■■■ Storage Location: 0002 Spare Parts ■ ■■■ PLANT: 2000 (San Diego) ■ ■■■ PURCHASING ORG: 1000 (Global) → table T024E ■ ■■■ Purchasing Group: 001 Pumps, 002 Electronics → T024 ■ ■■■ SALES ORG: 1000 (US West) → table TVKO ■ ■■■ Distribution Channel: 10 Direct, 20 Wholesale → TVTW ■ ■■■ Division: 01 Pumps, 02 Motors → TSPA ■ SALES AREA = SalesOrg 1000 + Channel 10 + Division 01 ■ ■■■ CREDIT CONTROL AREA: 1000 → table T014
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Config Object	Transaction	Table	ECC	S/4	Example
Define Company	OX15	T880	■	■	SEMP = Sempra Energy (holding)
Define Company Code	OX02	T001	■	■	1000 = Southern California Gas
Assign CoA to CoCode	OB62	T001	■	■	CoCode 1000 → USCA
Define Controlling Area	OKKP	TKA01	■	■	1000 = North America CO Area
Assign CoCode to CoArea	OX19	TKA02	■	■	CoCode 1000 → CoArea 1000
Define Plant	OX10	T001W	■	■	1000=Los Angeles, 2000=San Diego
Define Storage Location	OX09	T001L	■	■	0001=Main WH, 0002=Spare Parts
Assign Plant to CoCode	OX18	T001W	■	■	Plant 1000 → CoCode 1000
Define Purchasing Org	OX08	T024E	■	■	1000 = Global Procurement
Assign POrg to Plant	OX17	T024W	■	■	POrg 1000 → Plant 1000, 2000
Define Purchasing Group	OME4	T024	■	■	001=Pumps buyers, 002=Electronics
Define Sales Organisation	OVX5	TVKO	■	■	1000=US West, 2000=US East
Define Distribution Channel	OVXI	TVTW	■	■	10=Direct, 20=Wholesale
Define Division	OVXB	TSPA	■	■	01=Pumps, 02=Motors, 03=Services
Assign Sales Org to CoCode	OVX3	TVKO	■	■	SOrg 1000 → CoCode 1000
Assign Plant to Sales Org	OVX6	TVKWZ	■	■	Plant 1000 → SOrg 1000
Define Valuation Area	OMWM	T001K	■	■	= Plant in standard setup
Define Warehouse Number	OMB1	T300	■	■	001=LA Main Warehouse
Assign WH to Plant/SLoc	OMB2	T320	■	■	WH001 → Plant 1000, SLoc 0001
Credit Control Area	OB45	T014	■	■	1000=US Credit Control Area

05 FI — Accounts Receivable Configuration

Customer groups, payment terms, dunning, credit management

Config Object	Transaction	Table	ECC	S/4	Example
Customer Account Groups	OBD2	T077D	■	■	KUNA=general, CPD=one-time, INTE=internal
Customer Number Ranges	XDN1	NRIV	■	■	External 1000000–1999999
BP Number Ranges	BUCF	SNRO	■	■	BP group CUST → range 1000000–1999999
BP Groupings	BUCF	BU_PARTNERGR	■	■	CUST, VEND, CPD, INTE
BP Roles	BUCF	BUT_ROLES	■	■	FLCU00=Cust General, FLCU01=Cust CoCode
CVI: Assign Cust Roles	SPRO→CVI	CVI_CUST_LINK_DEF	■	■	KUNA → FLCU00+FLCU01
Payment Terms	OBB8	T052	■	■	Z030=Net 30, Z014=14 days 2% discount
Dunning Procedures	FBMP	T047	■	■	Z001=3-level dunning
Dunning Levels	FBMP	T047S	■	■	Level 1=7d, 2=14d, 3=30d past due
Dunning Texts	FBMP	T047T	■	■	Letter text per dunning level
Interest Calculation	OB46	T056	■	■	8% annual on overdue items
Reconciliation Account	FS00	SKB1	■	■	G/L 140000 = A/R Recon (type D)
Tolerance Groups (Cust.)	OBA3	T043T	■	■	Max diff: $5 or 0.5% (whichever less)
Credit Control Area	OB45	T014	■	■	1000 = US Credit Control
Credit Mgmt (ECC)	FD32	KNKK	■	■	Per-customer credit limit (old)
Credit Mgmt (S/4 FSCM)	UKM_BP	UKM_BP_ATTR	■	■	Score-based, real-time, per BP
Output (Invoice PDF)	V/40	TNAPR	■	■	RD00=invoice output via email/print

Example — Payment Terms Z030: Customer Siemens AG orders $10,000 of pumps. Invoice FB70 raised. Payment term Z030 = Net 30 days. Due date = invoice date + 30 days. If unpaid → dunning level 1 letter at day 37 → level 2 at day 44 → level 3 (legal) at day 60. In S/4: credit check via UKM_BP blocks new orders if Siemens exceeds $500,000 credit limit.

06 FI — Accounts Payable Configuration

Vendor groups, payment program, house banks, withholding tax

Config Object	Transaction	Table	ECC	S/4	Example
Vendor Account Groups	OBD3	T077K	■	■	0001=domestic, 0002=foreign, CPD=one-time
Vendor Number Ranges	XKN1	NRIV	■	■	External 2000000–2999999
BP Roles (Vendor)	BUCF	BUT_ROLES	■	■	FLVN00=Vend General, FLVN01=Vend CoCode
CVI: Assign Vend Roles	SPRO→CVI	CVI_VEND_LINK_DEF	■	■	0001 → FLVN00+FLVN01
Payment Terms	OBB8	T052	■	■	Z014=14 days 2% discount, Z000=immediate
Payment Methods	FBZP	T042Z	■	■	T=Bank transfer, C=Cheque, B=Bill
House Banks	FI12	T012	■	■	MAIN=Citibank LA, WEST=WellsFargo
Bank Accounts	FI12	T012K	■	■	113100=Citibank checking account
Bank Master (IBAN)	FI01	BNKA	■	■	Bank BIC, name, address
Vendor Bank (S/4)	BP→IBAN	TIBAN	■	■	IBAN replaces LFBK account number
Payment Program	FBZP	T042	■	■	F110 run: propose → display → pay
Paying Company Codes	FBZP	T042B	■	■	CoCode 1000 pays for CoCode 2000 too
Payment Run (F110)	F110	REGUH/REGUP	■	■	Batch: select due items → create pmts
WHT Types	SPRO→FI→WHT	T059P	■	■	W1=1099-MISC, W9=1099-DIV
WHT Codes	SPRO	T059Z	■	■	W1-01=backup WHT 24%, W1-02=standard 0%
Reconciliation Account	FS00	SKB1	■	■	G/L 210000 = A/P Recon (type K)
Check Double Invoice	SPRO	T001	■	■	XDEZV=X: block duplicate ref+vendor+amt

Example — F110 Payment Run: Vendor SAP SE invoice $45,000, due in 14 days, payment method T (transfer), house bank MAIN. F110 run: identifies all due items → creates payment proposal → accounting posts DR Vendor 210000 $45,000 / CR Bank 113100 $45,000 → ACDOCA entry created → DME file sent to Citibank → IBAN DE89 3704 0044 debited.

07 FI — Asset Accounting (New AA in S/4HANA)

Chart of depreciation, asset classes, depreciation keys, new engine

S/4HANA introduces **New Asset Accounting (New AA)**. The biggest change: depreciation is no longer a batch job (AFAB) — it is calculated **on-the-fly per posting** and stored directly in ACDOCA. Parallel depreciation for IFRS, US GAAP and Tax simultaneously.

Config Object	Transaction	Table	ECC	S/4	Example
Chart of Depreciation	EC08	T090	■	■	US00=US GAAP, IAS=IFRS, DE=German HGB
Asset Classes	OAOA	ANKA	■	■	1000=Land, 2000=Buildings, 3000=Machinery
Asset Class Number Ranges	AS08	NRIV	■	■	Class 3000 → assets 3000000–3999999
Depreciation Areas	OADB	T090R	■	■	01=Book (IFRS), 15=Tax, 32=Local GAAP
Depreciation Keys	AFAMA	TAFA	■	■	LINR=straight line, DG10=10% declining bal
Useful Life per Asset Class	OAYZ	T090N	■	■	Class 3000 → 10 years useful life
G/L Account Assignment	AO90	FIAA_ACCOUNT	■	■	Asset class 3000 → Depr exp 850000
Screen Layout (Asset)	AO73	T015	■	■	Which fields required per asset class
Transaction Types	OAYA	TTTYP	■	■	100=acquisition, 200=retirement, 600=depr
New Depreciation Engine	SPRO	FAAT_DOC_TYPE	■	■	S/4: calculates from asset value date
Asset Values in ACDOCA	Auto	ACDOCA	■	■	All asset postings in universal journal
Parallel Ledger Depr.	OADB	T090R	Opt	■	IFRS area 01 + Tax area 15 simultaneously

Example — Asset Pump-Station-001 (class 3000, LINR): Acquired 01.01.2026 for $120,000. Useful life = 10 years. Annual depreciation = $12,000. ECC: AFAB batch job runs monthly → posts $1,000/month. S/4 New AA: depreciation calculated at time of posting → ACDOCA: DR Depr Exp 850000 $1,000 / CR Accum Depr 175000 $1,000 — automatically for IFRS area AND Tax area simultaneously.

08 MM — Materials Management Configuration

Account determination, material types, movement types, Material Ledger

Account Determination — How Stock Movements Post to G/L

This is the most critical MM config. Transaction **OBYC** maps: movement type + valuation class → G/L account automatically. No manual G/L entry needed.

Transaction Key	Meaning	Dr Account	Cr Account	Example
BSX	Inventory posting	300000 Raw Mat / 310000 FG		GR: DR Inventory
WRX	GR/IR clearing	—	191100 GR/IR	GR: CR GR/IR
GBB+VAX	Goods issue (sales)	893000 COGS	—	Delivery: DR COGS
GBB+VBR	GI to cost centre	400000 Expense	—	Maintenance issue
GBB+AUF	GI to prod order	WIP account	—	Issue components
PRD	Price difference	281000 Variance	—	MAP recalculation
AUM	Stock transfer	Dest inventory	Src inventory	Inter-plant transfer
KBS	GR to cost centre	Cost ctr expense	—	Non-stock PO receipt

Config Object	Transaction	Table	ECC	S/4	Notes
Valuation Classes	OMSK	T025, T025T	■	■	7920=FG, 3000=RM, 7900=HALB
Account Category Reference	OMSK	T025B	■	■	Links mat type to val class group
Account Determination	OBYC	T030	■	■	BSX+7920 → G/L 310000
Material Types	OMS2	T134, T134M	■	■	FERT, ROH, HALB, HAWA, DIEN
Material Groups	OMSF	T023, T023T	■	■	001=Pumps, 002=Motors
Movement Types	OMJJ	T156, T156T	■	■	101=GR PO, 601=GI delivery
Material Number Ranges	MMNR	NRIV	■	■	External 1–999999, Internal auto
Purchasing Value Keys	SPRO	T068	■	■	Default tolerances for materials
Tolerance Keys	SPRO	T169G	■	■	B1=order price, B2=delivery cost
Source List	ME01	EORD	■	■	Preferred vendor for material/plant
Info Records	ME11	EINA, EINE	■	■	Agreed price: vendor X, mat Y=$450
Activate Material Ledger	OMX1	CKMLMVOBJ	Opt	MANDATORY	Activate per plant — mandatory in S/4
ML Actual Costing	CKMVFM	CKMLHD	Opt	MANDATORY	Period-end actual cost calc
Material Document	Auto	MATDOC	■	■	Replaces MKPF+MSEG in S/4
Batch Management	SPRO	T001W	■	■	Per plant — cannot reverse once stock exists
Serial Number Profiles	SPRO	SERP	■	■	Track each unit individually

Key Movement Types with Accounting: 101 GR vs PO: DR Inventory 310000 / CR GR/IR 191100 (vendor goods arrive) 261 GI to Production: DR WIP / CR Inventory 310000 (components issued to order) 601 GI for Delivery: DR COGS 893000 / CR Inventory 310000 (pump shipped to customer) In S/4: all post to ACDOCA with material, plant, profit centre stamped on every line.

09 SD — Sales & Distribution Configuration

Sales doc types, pricing, output, partner determination, revenue account

Pricing — How a Customer Price is Calculated

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Customer orders PUMP-100, Sales Org 1000, Channel 10: Pricing Procedure RVAA01 runs — condition types in sequence: Step 10 PR00 List Price +$500.00 (from material price list) Step 20 K007 Customer Discount -$25.00 (5% for key account) Step 30 K029 Material Discount -$10.00 (2% special promotion) Step 40 MWST Output Tax +$37.35 (8.09% California) ■■■
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10 PP — Production Planning Configuration

BOM, routings, work centres, MRP, production orders

BOM + Routing + Work Centre = Standard Cost Estimate

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PUMP-100 Standard Cost Estimate: Bill of Materials (CS01 → STKO/STPO): Routing (CA01 → PLKO/PLPO):
SHAFT-01 1 EA × $80 = $80 Op 0010 Cut WC:LATHE-01 0.5 HR BODY-01 1 EA × $120 = $120 Op 0020 Weld
WC:WELD-01 1.0 HR SEAL-KIT 1 SET× $35 = $35 Op 0030 Test WC:TEST-01 0.5 HR PAINT-01 2 L × $5 = $10
Material Cost: $245 Labour 2HR × $40 = $80 Machine 2HR × $25 = $50 Overhead 25% on labour = $20 Total
Standard Cost: $245+$150 = $395 MBEW-STPRS = $395/EA
```

Config Object	Transaction	Table	ECC	S/4	Example
Work Centres	CR01	CRHD, CRCA	■	■	LATHE-01=turning, WELD-01=welding
Capacity Categories	OPJK	CR tables	■	■	Machine hrs, Labour hrs, Tool
Routings	CA01	PLKO, PLPO	■	■	PUMP-100: cut→weld→test operations
Bill of Materials	CS01	STKO, STPO	■	■	PUMP-100 = SHAFT+BODY+SEAL+PAINT
BOM Usage	SPRO	STZU	■	■	1=production, 5=sales, 3=universal
MRP Types	OMDT	T438M	■	■	PD=MRP, VB=reorder point, ND=no plan
Lot Sizes	OMI4	T438D	■	■	EX=exact, FX=fixed 100, HB=replenish max
Scheduling Parameters	OPUZ	T490	■	■	Backward scheduling, float times
Production Order Types	OPL8	T003O	■	■	PP01=standard, PP02=rework order
Costing Variants	OKKN	TKK4V	■	■	PPC1=planned cost, PPP1=actual cost
Results Analysis Keys	SPRO	TKKAV	■	■	WIP calculation method at month end
Variance Keys	OKV1	TKKVS	■	■	000001=full variance calculation
Overhead Cost Sheet	KZS2	TKKOS	■	■	25% overhead surcharge on labour
Production Versions	SPRO	MKAL	■	■	0001=BOM alt 1+Routing 1
MRP Live (S/4)	SPRO	PPH_* tables	■	■	Real-time MRP — no overnight batch
Demand Management	MD61	PBIM	■	■	Planned independent requirements

11 CO — Controlling Configuration

Cost centres, profit centres, COPA, internal orders, ACDOCA merger

In S/4HANA, **CO is no longer separate from FI**. Both post to ACDOCA. The old reconciliation between FI and CO (transaction KALC) is no longer needed. Also: Cost Elements are gone as a separate object — every G/L account IS a cost element.

Config Object	Transaction	Table	ECC	S/4	Example / Change
Controlling Area	OKKP	TKA01	■	■	1000=North America, currency USD
Cost Element Types (ECC)	KA01	CSKU	■	■	ECC only — separate object, now gone
Cost Elements = G/L (S/4)	FS00	SKA1	■	■	Every G/L is auto a cost element
Cost Centre Types	OKA2	TKA04	■	■	A=production, B=admin, C=service
Cost Centres	KS01	CSKS, CSKT	■	■	1000=Pumps Prod, 2000=Finance
Cost Centre Hierarchy	OKEON	SETHEADER	■	■	Standard hierarchy for reporting
Profit Centres	KE51	CEPC, CEPCT	■	■	PC_PUMPS, PC_MOTORS, PC_SERVICES
Profit Centre Hierarchy	KCH1	SETHEADER	■	■	All PCs under one root node
Internal Orders	KO01	AUFK	■	■	Marketing campaign, R&D project
Order Types (CO)	KOT2	T003O	■	■	0100=overhead order, 0200=invest
Activity Types	KL01	CSLA	■	■	LABOUR=machine hours, SETUP=setup hrs
Statistical Key Figures	KK01	CSKF	■	■	HEADCNT=headcount, SQMETR=floor space
Assessment Cycles	KSU1	RKCT_* tables	■	■	IT cost → all plants by headcount
Distribution Cycles	KSV1	RKCA_* tables	■	■	HR cost → departments by headcount
COPA (Costing-based)	KEA0	CE4xxxx	■	Deprc	Old COPA — phased out in S/4
COPA (Account-based)	KEA0	ACDOCA	Opt	■	Preferred: same table as FI — no recon
Material Ledger	OMX1	CKMLHD	Opt	MADabe	Actual cost per period, mandatory S/4
Plan/Actual Comparison	KSBL	COSS/COSP(ECC)	■	ACDOCA	Plan vs actual in ACDOCA in S/4

COPA Example — Account-Based (S/4 preferred): Sale of PUMP-100 to Siemens AG, $502.35, COGS $395: ACDOCA lines: Revenue $502.35 → Profit Centre PC_PUMPS, Segment SEG_INDUSTRIAL, Customer 100045, Material PUMP-100, Plant 1000 COGS $395 → same dimensions. Gross Margin = $107.35 visible in REAL-TIME — no month-end COPA transfer needed.

12 QM — Quality Management Configuration

Inspection types, sampling, results recording, usage decisions

Config Object	Transaction	Table	Example
Inspection Types	SPRO	QPCA	01=GR, 04=production, 06=delivery, 10=audit
Sampling Procedures	QDV1	QDSV	100% check, AQL 0.65, fixed sample 10
Inspection Methods	QS21	QPMK	Dimensional check, visual inspection, lab test
Inspection Characteristics	QS21	QPMK	LENGTH 450mm±2mm, PRESSURE 16BAR
Defect Codes	QS41	QPCD	DENT=surface dent, CRACK=hairline crack
Catalogs	QS61	Q* tables	Usage decisions: A=accept, R=reject, I=block
Notification Types	SPRO	QMIH	Q1=internal, Q2=customer complaint, Q3=vendor
Quality Info Record	QI01	QINF	Vendor X certified for PUMP-100, skip lot 1/10
Inspection Lot (auto created)	MIGO/auto	QMEL	Created on GR if MARC-INSMK=1
Results Recording	QE01	QMRS	Inspector records measured values vs specs
Usage Decision	QA11	QMUR	A=accept → move 321 to unrestricted stock

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QM Flow Example: GR for PUMP-100 (MARC-INSMK=1) → stock to QI automatically Inspector: QE01 records LENGTH=449mm (spec 450±2mm) → PASS Inspector: QA11 usage decision A=Accept → Movement 321 → stock to unrestricted If REJECT: Movement 344 → blocked stock → Q2 notification → vendor return Movement 122
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13 PM — Plant Maintenance Configuration

Equipment, functional locations, maintenance orders, integration

Config Object	Transaction	Table	Example
Functional Locations	IL01	IFLOT	SEP-LA-PUMP-001 = Pump station Los Angeles
Equipment Master	IE01	EQUI	PUMP-001 S/N A12345 installed at SEP-LA-PUMP-001
Maintenance Plant	SPRO	T001W	1000 = Los Angeles maintenance plant
Order Types	OIOA	T003P	PM01=corrective, PM02=preventive, PM03=inspection
Work Centre (PM)	IR01	CRHD	MECH-01=Mechanics, ELEC-01=Electrical
Catalog (PM)	QS41	QPCD	Breakdown codes: ELEC=electrical, MECH=mechanical
Maintenance Strategies	IP11	MMPT	Annual service, Quarterly check, 5000hr service
Task Lists (PM)	IA05	PLKO	Standard ops: inspect bearings, change oil, test
Planner Groups	SPRO	T024I	001=Mechanical, 002=Electrical, 003=Civil
Notification Types	SPRO	QMIH	M1=malfunction, M2=maintenance request
Settlement Rules	SPRO	TKA3A	PM Order cost → Cost Centre or Asset

```
PM Integration Flow: Equipment PUMP-001 breaks → IW21 Notification M1 (malfunction) → IW31 PM Order
PM01 created (corrective) → Spare parts reserved (MM: SEAL-KIT-01 from SLoc 0002) → Labour assigned:
MECH-01 work centre, 4 hours → Parts issued: MVT 261 → ACDOCA: DR PM Order / CR Inventory 200000 →
Labour confirmed: CO → ACDOCA: DR PM Order / CR Cost Centre MECH → Order settlement: total cost $850 →
Cost Centre 1000 (ACDOCA) → If capital repair > threshold: capitalise to Asset → FI-AA depreciation
```

14 HR/HCM — Human Resources Configuration

Org structure, payroll, infotypes, FI integration

Config Object	Transaction	Table	Example
Personnel Areas	SPRO	T500P	1000=Los Angeles, 2000=San Diego
Personnel Subareas	SPRO	T503	0001=Office staff, 0002=Plant workers
Employee Groups	SPRO	T501	1=Active, 2=Retired, 3=Contractor
Employee Subgroups	SPRO	T503K	D1=Hourly, D2=Salaried, D3=Executive
Pay Scale Types	SPRO	T510A	US01=US hourly, US02=US salaried
Pay Scale Levels	SPRO	T510	Level 1-10 per pay scale type
Wage Types	SPRO	T512T	/001=basic pay, /006=overtime 1.5x, /550=bonus
Payroll Areas	SPRO	T549A	US=US monthly, BI=biweekly
Payroll Schema	PE01	T52C0	US schema: gross→tax→deductions→net pay
Cost Centre Assignment	PP01	HRP1001	Employee 10001 → Cost Centre 1000
Infotype 0001 (Org)	PA30	PA0001	Employee, position, org unit, cost centre
Infotype 0002 (Personal)	PA30	PA0002	Name, DOB, nationality, ID
Infotype 0008 (Basic Pay)	PA30	PA0008	Salary $85,000/year, pay scale D2-L5
Infotype 0009 (Bank)	PA30	PA0009	Employee IBAN for salary payment
Infotype 0071 (Pension)	PA30	PA0071	401k: employee 6%, employer match 3%

```
Payroll → FI Integration: Monthly payroll run (PC00_M99_CIPE) for 500 employees: ACDOCA posting: DR Labour Cost 500000 $2,500,000 (reads IT0001 cost centre per employee) CR Federal Tax Payable $ 500,000 (reads IT0071 tax data) CR Social Security Pay. $ 190,000 CR Payroll Clearing $1,810,000 (net pay to employees) Next step: F110 payment run → IBAN per employee (IT0009) → bank transfer
```

15 TR — Treasury Configuration

House banks, IBAN, payment factory, cash management

Config Object	Transaction	Table	ECC	S/4	Example
House Banks	FI12	T012	■	■	MAIN=Citibank LA, WEST=WellsFargo Phoenix
Bank Accounts	FI12	T012K	■	■	MAIN-CHK=checking 113100, MAIN-SAV=savings
Bank Master	FI01	BNKA	■	■	Bank BIC CITIUS33, name, address
Vendor IBAN (S/4)	BP	TIBAN	■	■	DE89 3704 0044 0532 0130 00
Customer IBAN (S/4)	BP	TIBAN	■	■	GB29 NWBK 6016 1331 9268 19
Bank Communication Mgmt	BCM	BCM tables	Opt	■	SWIFT MT100, XML PAIN.001
Cash Journal	FBCJ	CSKS	■	■	Petty cash transactions, daily balance
Cash Management (ECC)	FF7A	FDFD	■	■	Separate transaction — legacy
Cash Mgmt (S/4 embedded)	FF7A	ACDOCA	■	■	Real-time cash position from ACDOCA
Electronic Bank Statement	FF_5	FEBEP	■	■	MT940 bank statement auto-reconcile
Liquidity Planner	LQ01	FLQMG	Opt	■	30-day forward cash forecast
Payment Factory	FBZP	T042	■	■	Central payments for all group entities

16 Module Integration Map — How Everything Talks

Every module connection, trigger, and ACDOCA posting

From Module	Event / Trigger	To Module	Result / ACDOCA Posting
MM / MRP	Stock below reorder point or sales order	PP	Demand → Planned Order created → Production Order
MM	Goods Receipt against PO (MVT 101)	FI	DR Inventory 310000 / CR GR/IR 191100 (ACDOCA)
MM	Invoice Verification (MIRO)	FI	DR GR/IR 191100 / CR Vendor 210000 (ACDOCA)
MM	Goods Issue to Production (MVT 261)	FI+CO	DR WIP / CR Inventory; cost on PP Order (ACDOCA)
MM	Goods Issue for Delivery (MVT 601)	FI+CO	DR COGS 893000 / CR Inventory 310000 (ACDOCA)
SD	Sales Order created (VA01)	MM+PP	ATP check vs MARD; MRP demand if insufficient
SD	Billing document (VF01)	FI	DR Customer 140000 / CR Revenue 400000 (ACDOCA)
SD	Credit check on order	TR/FI	UKM_BP checks BP credit score → block if over limit
PP	Production Order GR (MVT 101)	MM+FI	Finished goods added to MARD; ACDOCA: DR FG / CR WIP
PP	Production Order settlement	CO+FI	Variance: DR/CR ACDOCA; cost to Profit Centre
PM	Spare parts issued to maint. order	MM+FI	MVT 261: DR PM Order / CR Inventory (ACDOCA)
PM	Order settlement	CO+FI	Maintenance cost → Cost Centre or Asset (ACDOCA)
HR	Payroll posting	FI+CO	DR Labour (per cost centre) / CR Payroll Clearing (ACDOCA)
AA	Depreciation posting (New AA)	FI	DR Depr Expense 850000 / CR Accum Depr 175000 (ACDOCA)
AA	Asset acquisition (capitalise)	FI+CO	DR Asset / CR Clearing or Vendor (ACDOCA)
QM	Usage decision REJECT (MVT 344)	MM+SD	Stock blocked; Q2 notification; vendor return MVT 122
CO	Period-end allocation (assessment)	FI+CO	IT cost → plants by headcount (ACDOCA reposting)
ML (CO)	Period-end actual cost calc	FI+CO+MM	Revalue inventory + COGS to actual (ACDOCA)

17 ACDOCA — The Universal Journal (S/4HANA Only)

Every financial posting lands here — FI, CO, MM, SD, AA, ML all in one table

ACDOCA = **AC**counting **DOC**ument **A**ctual. This is the single most important table in S/4HANA. In ECC you needed BKPF+BSEG (FI) + COEP (CO) + ANLC (AA) + CE4x (COPA) — 5+ tables, all needing reconciliation. In S/4HANA: **one table, always reconciled, real-time.**

ACDOCA Field Group	Field	Description	Example Value
Identity	RCLNT / RBUKRS	Client / Company Code	100 / 1000
Identity	GJAHR / BELNR	Fiscal Year / Document Number	2026 / 5000012345
Identity	DOCLN / BLDAT	Line Number / Document Date	001 / 15.10.2026
G/L	RACCT	G/L Account	310000 (FG Inventory)
G/L	DRCRK / HSL	Dr/Cr Indicator / Amount Local	S (Debit) / 45,000.00
G/L	WSL / TSL	Transact. Currency / 2nd Local	45,000.00 USD / 41,400.00 EUR
Controlling	RCNTR	Cost Centre	1000 (Pumps Production)
Controlling	PRCTR	Profit Centre	PC_PUMPS
Controlling	SEGMENT	Segment	SEG_INDUSTRIAL
Controlling	RFAREA	Functional Area	PRODUCTION
Logistics	MATNR / WERKS	Material / Plant	PUMP-100 / 1000
Logistics	KUNNR	Customer	100045 (Siemens AG)
Logistics	LIFNR	Vendor	200078 (SAP SE)
Logistics	VBELN	Sales Document	2000001234
Logistics	EBELN	Purchase Document	4500012345
Logistics	AUFNR	Order (PP/PM/CO)	1000001234 (Production Order)
Asset	ANLN1 / ANLGR	Asset Number / Group	100000 / MACHINERY
Quantity	MENGE / MEINS	Quantity / Unit	100 / EA

ACDOCA Power Query Example: Show me gross margin by Profit Centre for October 2026: SELECT PRCTR, SUM(CASE WHEN RACCT LIKE '4%' THEN HSL ELSE 0 END) AS Revenue, SUM(CASE WHEN RACCT='893000' THEN HSL ELSE 0 END) AS COGS FROM ACDOCA WHERE GJAHR=2026 AND POPER=10 AND RBUKRS='1000' GROUP BY PRCTR In ECC: needed FAGLFLEXT (FI) + COPA extraction + CO reconciliation report — 3 steps. In S/4: ONE query, real-time, zero reconciliation.

18 Complete Table Reference — ECC vs S/4HANA

Every important table — what changed, what stayed, what is new

Tables That CHANGED or DISAPPEARED in S/4HANA

Module	ECC Table	S/4HANA	Status	Impact / Action Required
FI	BKPF	ACDOCA (view)	Compatibility view	Custom programs using BKPF still work — read-only
FI	BSEG	ACDOCA (view)	Compatibility view	Custom reports must be rewritten for ACDOCA
FI	GLT0	Calculated	REMOVED	No totals table — balance calculated live from ACDOCA
FI	FAGLFLEXT	Calculated	REMOVED	New G/L totals gone — use ACDOCA queries
FI	FAGLFLEXA	ACDOCA	MERGED	All FI line items now in ACDOCA
CO	COEP	ACDOCA	MERGED	CO and FI in same table — no reconciliation
CO	COSS/COSP	Calculated	REMOVED	CO object totals calculated from ACDOCA
CO	CE4xxxx	ACDOCA	Deprecated	Costing-based COPA phased out — use account-based
MM	MKPF	MATDOC	REPLACED	Material document header → new MATDOC table
MM	MSEG	MATDOC	REPLACED	Material document items → new MATDOC table
MM	RBKP	ACDOCA	REPLACED	Invoice Verification header → ACDOCA
MM	RSEG	ACDOCA	REPLACED	Invoice Verification items → ACDOCA
SD	KONV	PRCD_ELEMENTS	REPLACED	Pricing conditions → new faster table
FI-AA	ANLC	ACDOCA	REPLACED	Asset periodic values → universal journal
FI-AA	ANLZ	ACDOCA (values)	Extended	Time-dep. data kept; values in ACDOCA
BP	—	BUT000/BUT100	NEW	Business Partner replaces split KNA1/LFA1
ML	CKMLCR	CKMLCR+ACDOCA	MANDATORY	Material Ledger now always active

Tables That STAYED THE SAME

Module	Table	Description	Any Change?
MM	MARA	Material Master — General Data	Extended (40-char number optional)
MM	MARC	Material Master — Plant Data (MRP, Purchasing)	No change
MM	MBEW	Material Valuation (standard price, MAP)	Extended with ML fields
MM	MARD	Storage Location Stock	No change — still current
MM	MAKT	Material Descriptions (multi-language)	No change
MM	EKKO/EKPO	Purchase Order Header/Items	No change
SD	VBAK/VBAP	Sales Order Header/Items	No change
SD	VBRK/VBRP	Billing Document Header/Items	No change
SD	LIKP/LIPS	Delivery Header/Items	No change
PP	STKO/STPO	Bill of Materials Header/Items	No change

Module	Table	Description	Any Change?
PP	PLKO/PLPO	Routing Header/Operations	No change
PP	AUFK	Production/PM/CO Order Master	No change
HR	PA0001–PA9999	HR Infotypes	No change
FI	T001	Company Codes	No change
FI	T052	Payment Terms	No change
ALL	NRIV	Number Ranges	No change
ALL	T003	Document Types	No change
CO	CSKS/CEPC	Cost Centre / Profit Centre	No change
MM	EINA/EINE	Purchasing Info Records	No change
MM	EORD	Purchasing Source List	No change

19 Migration Impact by Module

Complexity rating, key changes, and what your team must do

Module	Complexity	Key Change in S/4	What Your Team Must Do
FI — G/L	■ HIGH	BSEG→ACDOCA, New G/L mandatory, doc splitting mandatory	Activate ledgers, configure splitting rules, rewrite custom G/L reports for ACDOCA
FI — A/R	■ MED	Customer→BP (CVI), credit mgmt FSCM	Migrate customers as BP, configure UKM_BP, re-test dunning and payment terms
FI — A/P	■ MED	Vendor→BP (CVI), IBAN mandatory	Migrate vendors as BP, convert LFBK→TIBAN, verify F110 payment run
FI — AA	■ HIGH	New depreciation engine, values in ACDOCA	Parallel run old vs new AA, validate asset values, migrate opening balances
CO	■ HIGH	COEP→ACDOCA, cost elements→G/L, account-based COPA preferred	Remove CE1x cost elements, switch to account COPA, activate ML for all plants
MM	■ MED	MSEG→MATDOC, ML mandatory, material number 40 chars optional	Migrate material masters, reconcile stock values, activate ML per plant
SD	■ MED	Customer→BP, KONV→PRCD_ELEMENTS, credit mgmt to FSCM	Re-test pricing, output, partner det., verify revenue account determination
PP	■ LOW	MRP Live optional, mostly same	Re-test MRP runs, validate production orders, cost estimates with new ML
PM	■ LOW	Mostly same, asset capitalisation via new AA	Re-test maintenance orders, verify cost settlement to cost centres/assets
QM	■ LOW	Mostly same	Re-test inspection lots, usage decisions, verify stock movement postings
HR	■ MED	Payroll → ACDOCA (was separate)	Verify payroll FI posting, test cost centre assignment, validate IBAN payments
TR	■ MED	IBAN mandatory, cash mgmt embedded	Convert all bank accounts to IBAN, test F110 payment, verify bank statements
WM→EWM	■ HIGH (if upgrading)	New architecture, new transactions	Separate EWM implementation workstream, new bin structure, new processes

20 End-to-End Build Sequence — Month by Month

From blank system to go-live — the correct implementation order

Phase	Month	Activity	Key Config / Tables	Owner
FOUNDATION	1	Activate S/4 system, client settings	T000, T005, T002, TCURC, T006	Basis
FOUNDATION	1	Chart of Accounts + G/L accounts	T004, SKA1, SKB1, T077S	FI
FOUNDATION	1	Company Codes + fiscal year	T001, T009, OB52	FI
FOUNDATION	1	Activate New G/L + ledgers + doc splitting	FINSC_LEDGER, FAGL_SPLINFO	FI
FOUNDATION	2	Controlling Area + cost elements	TKA01, SKA1 (= cost elements in S/4)	CO
FOUNDATION	2	Plants, storage locations	T001W, T001L	MM/Basis
FOUNDATION	2	Sales Orgs, Channels, Divisions	TVKO, TVTW, TSPA	SD
FOUNDATION	2	Purchasing Orgs + Groups	T024E, T024	MM
MODULE CFG	3	FI-AR: customer groups, payment terms, dunning	T077D, T052, T047	FI
MODULE CFG	3	FI-AP: vendor groups, payment program, banks	T077K, T042, T012	FI
MODULE CFG	3	FI-AA: asset classes, depreciation keys	T090, ANKA, TAFA	FI-AA
MODULE CFG	3	MM: material types, valuation classes, OBYC	T134, T025, T030	MM
MODULE CFG	4	MM: Activate Material Ledger all plants	CKMLMVOBJ, OMX1	CO/MM
MODULE CFG	4	SD: doc types, pricing, output, partner det	TVAK, T683, TNAPR	SD
MODULE CFG	4	PP: work centres, MRP types, lot sizes	CRHD, T438M, T438D	PP
MODULE CFG	4	CO: cost centres, profit centres, COPA	CSKS, CEPC, ACDOCA	CO
INTEGRATION	5	CVI: BP roles, number ranges, field mapping	BUT_ROLES, CVI_MAP_*	All
INTEGRATION	5	Account determination: OBYC, VKOA, AC03	T030, FIAA_ACCOUNT	FI/MM/SD
INTEGRATION	5	Credit management: UKM_BP config	UKM_PROF, UKM_BP_ATTR	FI/SD/TR
INTEGRATION	6	End-to-end process testing (all modules)	All	All teams
MASTER DATA	7	G/L Master, Cost Centres, Profit Centres	SKA1, CSKS, CEPC	FI/CO
MASTER DATA	7-8	Business Partners (Customers + Vendors)	BUT000, KNA1, LFA1, TIBAN	All
MASTER DATA	8-9	Material Master (MARA, MARC, MBEW, MVKE)	MARA, MARC, MBEW, MVKE	MM
MASTER DATA	9	BOM, Routings, Work Centres	STKO, PLKO, CRHD	PP
MASTER DATA	9	Asset Master + opening values	ANLA, ANLB, ACDOCA	FI-AA
OPEN ITEMS	10	Open POs, Sales Orders, Production Orders	EKKO, VBAK, AUFK	All
OPEN ITEMS	11	FI Open Items (AR+AP invoices)	ACDOCA opening entries	FI
OPEN ITEMS	11	Stock quantities → reconcile to FI G/L	MARD, MBEW, ACDOCA	MM/FI
CUTOVER	12	Parallel run: ECC and S/4 side by side	All	All
CUTOVER	12	Cutover weekend: freeze ECC, load final delta	All tables	All
GO-LIVE	12	Go-Live + hypercare (2 weeks daily BUPT)	ACDOCA, BUPT, MB5L	All

The Three Non-Negotiable Go-Live Gates: 1. ACDOCA balances match ECC legacy system balances (FI reconciliation) 2. MARD stock quantity × MBEW price = G/L inventory account balance (stock reconciliation) 3. BUPT (BP consistency) = ZERO errors for all Business Partners (master data quality) All three must be GREEN before cutover weekend. No exceptions.